TP02(MISorakmony)

Link: https://github.com/Mookneii/Data_Structure_And_Algorithms.git

1.

```cpp
#include <iostream>
using namespace std;
int sumOdd(int n){
    if(n<=0)
    return 0;
    if (n%2 != 0) return n + sumOdd(n-1);
    return sumOdd(n-1);
}
int main(){
    int n;
    cout<<"Enter n: ";
    cin>>n;
    cout<<"Sum of odd numbers = "<<sumOdd(n);
        return 0;
}
```

```
Enter n: 34
Sum of odd numbers = 289
```

2.

```cpp
#include <iostream>
using namespace std;

// a. Power function: m^n
int power(int m, int n) {
    if (n == 0)
        return 1;
    return m * power(m, n - 1);
}

// b. Sum of squares: 1^2 + 2^2 + ... + n^2
int sumSquare(int n) {
    if (n == 1)
        return 1;
    return n * n + sumSquare(n - 1);
}

// c. Sum digits of a number
int sumDigit(int n) {
    if (n == 0)
        return 0;
    return n % 10 + sumDigit(n / 10);
}

int main() {
    cout << "power(2,3) = " << power(2,3) << endl;
    cout << "sumSquare(4) = " << sumSquare(4) << endl;
    cout << "sumDigit(123) = " << sumDigit(123) << endl;

    return 0;
}
```

```
power(2,3) = 8
sumSquare(4) = 30
sumDigit(123) = 6
```

3.

```cpp
#include<iostream>
using namespace std;
void display(int n){
    if (n ==0){return;}
    cout<<n<<" ";
    display(n-1);
}
int main(){
    int n;
    cout<<"Enter n: ";cin>>n;
    cout<<"Display numbers from n to 1"<<endl;
    display(n);
}
```

```
Enter n: 43
Display numbers from n to 1
43 42 41 40 39 38 37 36 35 34 33 32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
```

4.

```cpp
#include <iostream>
using namespace std;
int findMin(int a[], int n){
    if(n ==1 ){
        return a[0];

    }
    int minRest = findMin(a, n-1);
    if (a[n-1] < minRest) return a[n-1];
    else{
        return minRest;
    }
}
int main(){
    int a[]={8,3,12,1,5};
    int n = 5;
    cout<<"Minimum element = "<<findMin(a, n);
    return 0;
}
```

```
Minimum element = 1
PS D:\Data Structure and Programming\Lab2>
```

5.

```cpp
#include <iostream>
using namespace std;
int findSum(int a[], int n){
    if(n ==0){
        return a[0];
    }
    return a[n-1]+ findSum(a, n-1);
}
int main(){

    int a[5]={12,5,7,21,6};
    cout<<"Sum = "<<findSum(a, 5);
    return 0;
}
```

```
Sum = 63
PS D:\Data Structure and Programming\Lab2>
```

6.

```cpp
#include <iostream>

using namespace std;

int askAndSum() {
    int num;
    cout << "Enter number : ";
    cin >> num;

    if (num == -1) {
        return 0;
    }

    return num + askAndSum();
}

int main() {
    int total = askAndSum();

    cout << "Total = " << total<< endl;
    return 0;
}
```

```
Enter number : 1
Enter number : 3
Enter number : 2
Enter number : 3
Enter number : -1
Total = 9
```